

Candidate Name

Centre Number

Candidate Number

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**

General Certificate of Education Advanced Level

CHEMISTRY**9189/5**

PAPER 5 Practical Test

JUNE 2013 SESSION

1 hour 20 minutes

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Insert

TIME 1 hour 20 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 6 printed pages and 2 blank pages.

Copyright: Zimbabwe School Examinations Council, J2013.

- 1 You are required to determine the concentration of ethanoic acid, CH_3COOH in gdm^{-3} in a sample of white vinegar, a food additive.

FA1 is a sample of white vinegar.

FA2 is 0.5 mol dm^{-3} aqueous sodium hydroxide, NaOH (aq) .

- (a) Pipette 25.00 cm^3 of FA2 into a conical flask and add a few drops of the indicator provided for the titration.

Titrate the contents of the conical flask with FA1 until the appropriate colour change is observed at the end point.

Repeat the titration as many times as you think necessary to obtain accurate results.

Record your burette readings in Table 1.

Table 1: Titration of FA1 with FA2

final burette reading / cm^3				
initial burette reading / cm^3				
volume of FA1 used				

[15]

Summary

25.00 cm^3 FA2 reacted with _____ cm^3 of FA1.

NB: Indicate clearly how you obtained this volume of FA1 from your results.

- (b) Name the indicator used.

[1]

- (c) Calculate the number of moles of sodium hydroxide, $\text{NaOH}_{(\text{aq})}$ which were placed in the conical flask.

For
Examiner's
Use

[1]

- (d) (i) Write a balanced chemical equation for the reaction taking place between FA1 and FA2.

[2]

- (ii) Hence calculate the number of moles of ethanoic acid in the volume of FA1 that reacted with FA2.

[1]

- (e) Calculate the concentration of ethanoic acid in gdm^{-3} in FA1.

$[\text{M}_r(\text{CH}_3\text{COOH}): 60]$

[2]

ASSESSMENT OF PLANNING SKILLS

DO NOT CARRY OUT YOUR PLAN

- (f) Another way of determining concentration of an acid is by titrating it with an alkali using a pH meter instead of an acid-base indicator.

You are required to give a plan of an experiment to determine the concentration of hydrochloric acid using the above method.

Assume you are provided with the following:

0.1 mol dm⁻³ NaOH (aq)
a sample of hydrochloric acid, HCl (aq)
a pH meter
common laboratory titration apparatus

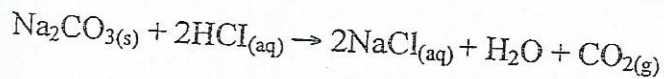
In your plan, state clearly a sequence of numbered steps you would take to perform the experiment and use your results to determine the concentration of the hydrochloric acid.

The Plan

FA3 is 2.00 mol dm^{-3} Hydrochloric acid, $\text{HCl}_{(\text{aq})}$.

FA4 is solid anhydrous sodium carbonate, $\text{Na}_2\text{CO}_{3(s)}$.

You are required to determine the enthalpy change, ΔH for the following reaction:



- (a) Weigh an empty weighing bottle. Place between 3.40g and 3.60g of FA4, solid anhydrous sodium carbonate in the weighing bottle and weigh it again.

Record your weighings in Table 2.1.

Table 2.1

mass of empty weighing bottle /g	
mass of weighing bottle + FA4 /g	
mass of weighing bottle + residual FA4 /g	
mass of FA4 added /g	

[3]

- (b) Using a measuring cylinder, place 60 cm^3 of FA3 into a plastic cup. Measure and record its temperature in Table 2.2.

Table 2.2

initial temperature of FA4 /°C	
temperature after mixing FA3 and FA4 /°C	
temperature rise /°C	

[5]

Tip the contents of the weighing bottle into the acid in the plastic bottle, stir gently with the thermometer and record the highest temperature obtained in Table 2.2.

Reweight the weighing bottle with any residual FA4 and record the mass in Table 2.1.

- (c) Assuming that 4.3 J are required to raise temperature of 1.0 cm^3 of the solution by 1°C , calculate the heat evolved.

[1]

For
Examiner's
Use

- (d) Calculate the number of moles of

(i) hydrochloric acid which were measured into the plastic cup,

[1]

(ii) sodium carbonate which were tipped into the plastic cup.

[$M_r \text{ Na}_2\text{CO}_3 : 106$]

[1]

- (e) Calculate the enthalpy change, ΔH , for the reaction between anhydrous sodium carbonate and hydrochloric acid.

[1]
[Total : 12]